

Stoichiometry

Stoichiometry is calculations involving the amounts of reactants and/or products in a chemical reaction.

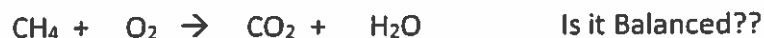
Stoichiometry calculations allow us to **relate** the number of particles in a chemical reaction to measurements of mass, volume and concentration.

BALANCED chemical equations tell us the number of moles (and particles) of each substance that are needed to react and that are produced. This is the MOLE RATIO.

That means, for example, if we needed to find out the amount of propane required to produce a specific amount of carbon dioxide, we could use the numerical coefficients.

The balanced chemical equation not only indicates chemicals present, their states (and maybe catalysts and whether the reaction is endothermic or exothermic if that information is included), but it also indicates the ratio of amounts involved in the reaction.

However, the balanced equation **cannot** reveal the speed of the reaction, any other specific conditions such as pressure and temperature, physical conditions (eg. solid vs powder, concentration of solutions, whether stirring or heating is required) or what is actually happening at the molecular level during chemical interactions.



Compound	methane	oxygen	carbon dioxide	water
Formula				
particles				
Molecules				
moles				

State two stoichiometric relationships from the above equation:

If you have 2 moles of methane reacting with excess oxygen, how many moles of carbon dioxide are produced?

Moles of water?